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Association between circulating concentrations of 25(OH)D and colorectal adenoma: a pooled analysis

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Abstract

The relationship between the biomarker of vitamin D status, 25(OH)D, and risk for colorectal neoplasia is suggestive but equivocal. Questions remain regarding whether there are differential associations between 25(OH)D and colorectal adenoma by gender, colorectal sub-site, or features of baseline and recurrent adenomas. We sought to investigate the relationship between 25(OH)D and both baseline and recurrent adenoma characteristics. This study was conducted among 2074 participants in a pooled population of two clinical intervention trials of colorectal adenoma recurrence. A cross-sectional analysis of 25(OH)D and baseline adenoma characteristics and a prospective study of recurrent adenomas and their characteristics were conducted. There was a statistically significant inverse association between 25(OH)D concentrations and the presence of three or more adenomas at baseline. Compared to participants with 25(OH)D levels <20 ng/ml, the adjusted ORs (95% CIs) were 0.99 (0.70–1.41) for those with concentrations of ≥20 and < 30 ng/ml, and 0.73 (0.50–1.06) among participants with levels ≥ 30 ng/ml (p-trend=0.05). Baseline villous histology was also significantly inversely related to 25(OH)D levels (p-trend=0.04). Conversely, 25(OH)D concentrations were not associated with overall colorectal adenoma recurrence, with ORs (95% CIs) of 0.91 (0.71–1.17) and 0.95 (0.73–1.24; p-trend = 0.85). These findings support the concept that the relationship between vitamin D and colorectal neoplasia may vary by stage of adenoma development.

Keywords

Vitamin D; colon cancer; 25(OH)D; colorectal adenoma

Introduction

Vitamin D has been intensively studied in relation to colorectal adenoma and cancer. Because it can be consumed through the diet or synthesized endogenously after UV exposure, the vitamin D metabolite 25-hydroxycholecalciferol [25(OH)D] is commonly employed as a marker for overall vitamin D status. Hydroxylation of 25(OH)D is required to produce the active vitamin D metabolite 1,25-dihydroxycholecalciferol [1,25(OH)₂D], which binds to the vitamin D receptor and modulates the transcription of numerous vitamin D target genes¹.

Epidemiological and clinical researchers have investigated the association between 25(OH)D and risk for colorectal adenoma, with some studies showing no overall relationship²⁻⁷. However, meta-analyses have revealed that low serum levels of 25(OH)D are significantly related to increased risk for both colorectal adenomas⁸⁻¹⁰ and cancer^{9, 11, 12}. The differences in results among individual studies may arise due to variation in the gender of the study participants^{13, 14}; colorectal sub-site investigated (distal vs. proximal)^{2, 13}; and study endpoint (incident adenoma compared to recurrent)⁸; however, few reports have included analyses of these characteristics within the same study population.

In addition to epidemiological data, much work has been conducted to identify the effect of vitamin D metabolites on colorectal carcinogenesis. Potential mechanisms of action that have been reported include anti-proliferative, pro-differentiating, and growth inhibitory effects of 1,25(OH)₂D^{15, 16}. This metabolite has also been demonstrated to inhibit β -catenin, a proto-oncogene that is a key modulator in colorectal carcinogenesis^{17, 18}.

Overall, vitamin D appears to be a promising candidate for chemoprevention and/or treatment of colorectal neoplasia. However, published data are relatively sparse for whether 25(OH)D is related to specific features of adenomas that render an individual to be at greater risk for colorectal cancer, and whether differences by gender or colorectal sub-site exist. Further, few studies to date have assessed whether 25(OH)D is related to both features of baseline adenomas as well as recurrent adenomas. Therefore, the objective of the present study was to investigate the associations between 25(OH)D and baseline and recurrent colorectal adenomas, as well as to evaluate whether these relationships vary in men compared to women; by distal or proximal location in the colon; or by advanced features of adenomas.

Materials and Methods

Study population

Data collected from the study populations of two randomized clinical trials, the Wheat Bran Fiber (WBF) Trial¹⁹ and the Ursodeoxycholic Acid (UDCA) Trial²⁰, were employed for the present study. As described elsewhere¹⁹, the objective of the WBF trial was to compare the effect of a high-fiber vs. a low-fiber cereal supplement on adenoma recurrence among individuals who had undergone colonoscopy and had one or more adenoma(s) removed. This randomized, double-blind, controlled trial was conducted with subjects recruited between 1990 and 1995 in Phoenix, Arizona; 1310 participants completed the study by undergoing one or more colonoscopies after randomization¹⁹, with a mean follow-up time from randomization to colonoscopy of 3.1 years. No differences in adenoma recurrence rates were observed between treatment groups¹⁹.

The UDCA trial design and participants were similar to the WBF Trial; it was a randomized, double-blind, placebo-controlled trial to compare the effect of UDCA on adenoma recurrence among patients that had a prior polyp removed at colonoscopy. Recruitment took place in Tucson and Phoenix, Arizona between 1995 and 1999²⁰. A total of 1192 participants completed the study after 3.2 years of follow-up²⁰, and the primary findings were that UDCA treatment had no effect on adenoma recurrence compared to the placebo group²⁰. All participants who completed these studies and who had available serum for analysis of 25(OH)D were eligible for the current study (n=2074).

Ethics

The WBF and UDCA studies were approved by the University of Arizona Human Subjects Committee and local hospital committees, and written informed consent was obtained from each participant prior to study enrollment.

Data Collection

Dietary, sociodemographic, and medical history data for all participants in the WBF and UDCA trials were collected at baseline with self-administered questionnaires. Dietary data were obtained with the Arizona Food Frequency Questionnaire (AFFQ), which was modified from the food frequency section of the National Cancer Institute's Health Habits and History Questionnaire²¹. The AFFQ is 113-item, semi-quantitative, scannable instrument, with study participants asked to report their usual intake of foods for the prior year²². A scale of seven categories ranging from >3 times/day to rarely/never was employed for most items; for beverages and commonly-consumed foods the scale ranged from >6 times/day to rarely/never²². Portion sizes of small, medium, or large were also recorded for each food item²². Calculation of total nutrient intake was performed by multiplying the frequency of each item's consumption by the nutrient composition for each portion size²².

Detailed data regarding adenoma characteristics such as number, size, location, and histology were collected for baseline adenomas and recurrent adenomas. As described previously, this information was obtained via medical records and pathology reports for each subject²⁰. The eligibility criteria for the WBF and UDCA trials required all participants to present with at least one baseline adenoma. After removal of baseline lesions, any colorectal adenoma detected at colonoscopy at least six months after randomization into either trial was counted as a recurrent adenoma. Adenomas were classified as advanced if they had a diameter of 1 cm or more, and/or tubulovillous or villous histology (at least 25% villous). Additionally, adenocarcinomas (n=12) were counted as advanced recurrences. All other adenomas were considered non-advanced. In subjects with more than one adenoma, size and characterization of the histologic type were based on the largest and/or most advanced adenoma.

Analysis of Vitamin D metabolite levels

Measurement of 25(OH)D concentrations was performed at Heartland Assays (Ames, IA) using a competitive chemiluminescence immunoassay²³. The laboratory utilized several QA/QC measures, including a pooled serum sample analyzed with batches of study samples to monitor analytical precision and identify possible laboratory shifts over time, as well as testing duplicates in different batches. The coefficient of variation was less than 7.0% for 25(OH)D analyses. All analyses were conducted in a blinded fashion. Participants were categorized by circulating concentrations of 25(OH)D into one of three categories: insufficient (< 20 ng/mL), sub-optimal (≥ 20 and < 30 ng/mL), and optimal (≥ 30 ng/mL)^{24, 25}.

Statistics

Descriptive data for baseline characteristics by study and by category of 25(OH)D concentration were calculated with means and standard deviations for the continuous variables and frequencies and percentages for the categorical variables. Unconditional logistic regression modeling was used to evaluate the associations between 25(OH)D concentrations and baseline adenoma characteristics, as well as for the relationship between 25(OH)D and adenoma recurrence. Variables assessed for potential confounding in both models were season of blood draw, intervention group, age, body mass index (BMI), gender, race, family history of colorectal cancer, current smoking, history of previous polyps, and aspirin use; dietary intake of fat, fiber, folate, magnesium, and calcium; supplement use; and energy intake. If a variable changed the point estimate by 10% or greater, it was included in the final multivariate logistic regression analyses of 25(OH)D and baseline adenoma characteristics and recurrence. The variables that met this criterion were body mass index, gender, age, race, and study and as such they were included in the final models. Data were complete for all variables with the exception of that for previous polyps; participants

missing these data were dropped from the descriptive data presented in Tables 1 and 2. A variable accounting for study (WBF vs. UDCA) was added to the final model to account for any potential differences between the two populations. Heterogeneity of effect between the WBF and UDCA studies was assessed by employing an interaction term for study and vitamin D status and evaluating with a likelihood-ratio test; a similar procedure was employed for heterogeneity of effect by gender. All analyses were conducted using STATA statistical software package [version 9.0, Stata Corporation, College Station, TX].

Results

Sufficient serum samples were available for measurement of 25(OH)D concentrations in 2074 (83.0%) of 2502 participants who completed the WBF and UDCA trials according to the study protocols described above. The mean time between blood draw and follow-up colonoscopy for the prospective analysis was approximately 3.1 years. Table 1 presents the characteristics of the study participants from the pooled population as well as the WBF and UDCA trials separately. The median age for all participants was 67 years, with an interquartile range (IQR) of 60.0–72.0. The median BMI (IQR) for the overall population was 27.3 (24.7–30.4). The study population was largely male (68.2%) and white, non-Hispanic (94.9%). A total of 457 subjects (22.0%) reported having a family history of colorectal cancer in one or more first-degree relatives, with 39.8% reporting a history of previous polyps. Of the participants, 12.7% were current smokers, and 28.0% reported having used aspirin in the month prior to study randomization. The age ($p<0.04$) and BMI ($p<0.01$) of participants in the UDCA trial were statistically significantly higher than for those in the WBF trial. There was a significantly greater proportion of participants in the UDCA trial who reported a family history of colorectal cancer (27.4%) or previous polyps (44.7%) as compared to the WBF trial (15.4% and 33.7%, respectively; p -values for both comparisons = 0.001).

Baseline characteristics of study participants by vitamin D status, as assessed by 25(OH)D concentrations, are described in Table 2. Lower BMI, male gender, and white race were all statistically significantly related to higher 25(OH)D levels, as was aspirin use in the past month. Dietary intake of fat, fiber, vitamin D, calcium, folate, and magnesium were directly and statistically significantly associated with 25(OH)D concentrations, as was energy intake; while use of dietary supplements was not related. Seasonal variation in 25(OH)D concentrations was observed, with mean levels of 26.3 ± 8.9 in the spring, 29.4 ± 10.7 in the summer, 29.7 ± 10.1 in the fall, and 24.9 ± 9.7 in the winter; however, inclusion of a variable for season in the logistic regression models did not materially alter the point estimates and therefore was not included in the models presented below.

Table 3 presents the adjusted odds ratios (95% CI) for the association between concentrations of 25(OH)D and characteristics of adenomas at baseline. In the pooled population, 25(OH)D levels were inversely related to the presence of three or more adenomas at baseline. Compared to participants with 25(OH)D concentrations of < 20 ng/ml, the ORs (95% CIs) were 0.99 (0.70–1.41) for those with concentrations of ≥ 20 and < 30 ng/ml, and 0.73 (0.50–1.06) among participants with 25(OH)D concentrations ≥ 30 ng/ml (p -trend=0.05). For adenoma size, no statistically significant relationship was observed for 25(OH)D in the pooled population, though in the UDCA trial there was a trend for reduced odds of large adenomas with increasing 25(OH)D concentration ($p=0.03$). There was a statistically significant inverse relationship between 25(OH)D and villous histology, with ORs (95% CIs) of 0.88 (0.66–1.18) and 0.72 (0.53–0.99; p -trend=0.04) in the pooled population. This finding was consistent within the WBF and UDCA trials when examined individually. There was no heterogeneity of effect between the two studies for the relationship between 25(OH)D and any of the baseline adenoma characteristics.

Additionally, no heterogeneity of effect was observed when the analyses were stratified by sex.

As shown in Table 4, there were no statistically significant associations between 25(OH)D and adenoma recurrence overall. Compared to those with 25(OH)D levels < 20 ng/ml, the OR (95% CI) for any recurrence in the pooled population was 0.91 (0.71–1.17) for participants with 25(OH)D concentrations of ≥ 20 and < 30 ng/ml; and 0.97 (0.75–1.26) for those with levels ≥ 30 ng/ml (p-trend= 0.95). However, a statistically significant direct relationship was observed for recurrence of ≤ 3 adenomas; with ORs (95% CIs) of 1.45 (0.96–2.20) for those with levels of ≥ 20 and < 30 ng/ml and 1.68 (1.09–2.58) for those with 25(OH)D concentrations ≥ 30 ng/ml (p-trend=0.02). No statistically significant relationships were observed between 25(OH)D and recurrent adenoma size or histology. As with the results for baseline adenomas, no heterogeneity of effect was observed between the UDCA and WBF studies, nor between sex-specific analyses. Table 5 presents the relationship between 25(OH)D concentrations and odds for baseline and recurrent adenomas by colorectal sub-site. No significant associations were found for 25(OH)D and either baseline or recurrent distal or proximal lesions.

Discussion

In the present study, a statistically significant inverse association was observed for concentrations of 25(OH)D and two features of baseline adenomas: the presence of three or more colorectal adenomas, and villous histology. In contrast, 25(OH)D levels were not associated with overall recurrence of colorectal adenomas or with recurrent adenoma size or histology. However, there was an unexpected, statistically significant, direct relationship between 25(OH)D concentrations and recurrence of ≥ 3 adenomas. No heterogeneity of effect was observed by study or gender for any of the analyses presented herein, nor were any differences by colorectal sub-site found.

To date, the association between measured concentrations of 25(OH)D and colorectal adenoma risk has remained suggestive, though equivocal. Several studies have reported significant inverse relationships between 25(OH)D and colon or colorectal adenoma^{26–30} and cancer^{31–38}, while other studies have reported null results^{2–7, 13, 14, 39, 40}. Meta-analyses of serum 25(OH)D and colorectal adenoma have uniformly shown a significant inverse relationship for adenoma incidence^{8, 10}, but not recurrence, though data for the latter outcome are sparse^{8, 10}. The present results are concordant with reported findings for recurrence, with no association between measured 25(OH)D and overall colorectal adenoma recurrence observed, but there was significant direct relationship between the vitamin D metabolite and recurrence of ≥ 3 adenomas.

Several studies have investigated whether vitamin D is related to specific adenoma characteristics or to advanced neoplasia, and results have been mixed. Fedirko et al.²⁸ conducted a comprehensive pooled analysis of the relationship between 25(OH)D and incident adenoma characteristics, including multiplicity, size, location, histology, and atypia, and observed no significant differences in the strength of the association between 25(OH)D and specific adenoma features. However, there was a suggestion of an inverse relationship between 25(OH)D and increasing number of incident adenomas (p=0.06)²⁸; these results are similar to the present analysis of 25(OH)D and baseline adenoma features, which demonstrated a statistically significant inverse association between 25(OH)D concentrations and number of adenomas present. In contrast, we observed an unexpected result for adenoma recurrence, whereby a statistically significant increased risk for ≥ 3 recurrent adenomas among those with higher 25(OH)D concentrations was found.

It is unclear why 25(OH)D may exhibit differential relationships with features of baseline and recurrent adenomas; particularly as data for associations with adenoma recurrence are relatively sparse^{8, 10}. Studies of cancer cell lines have in general revealed anti-proliferative, pro-differentiating effects of 1,25(OH)₂D 15, 16, with specific growth inhibitory action in colorectal cancer cell lines⁴¹. In addition, 1,25(OH)₂D and its analogs have been shown to induce G₀/G₁ cell-cycle arrest in cancer cell lines⁴², as well as to stimulate transcription of the tumor suppressor gene E-cadherin¹⁷ and reduce the transcriptional activity of β -catenin, a proto-oncogene that is implicated in several stages of colorectal carcinogenesis¹⁷. Higher plasma levels of 25(OH)D have also been associated with decreased proliferation in tissue from subjects at risk for colorectal neoplasms⁴³, a finding that is of particular interest because of reports indicating that colon carcinoma (Caco-2) cells display 25-hydroxyvitamin D₃ 1-alpha hydroxylase (CYP27B1) activity⁴⁴, and as such can produce 1,25(OH)₂D in colon tissue independently of the classic metabolic pathway in the kidney. Expression of CYP27B1 has been reported to be higher in colon adenomas than in normal colonic cells⁴⁵, and remains high until tumors become undifferentiated⁴⁶.

One speculative explanation regarding the potential mechanism for the unexpected result of a statistically significant increased risk for ≥ 3 recurrent adenomas among those with higher 25(OH)D concentrations may be due to differential activity of 25(OH)D by stage of adenoma development. Several studies, as reviewed by Lao et al.⁴⁷, have reported significant epigenetic and methylation pattern changes during the histological stages of the colorectal cancer polyp- to- adenocarcinoma transition. These epigenetic conversions could have profound effects on the bioactions of vitamin D metabolites in adenoma tissue that might vary by stage of adenoma development. Moreover, anabolic CYP27B1 expression, as well as catabolic CYP24A1 expression, can fluctuate in a stage-dependent fashion in adenoma tissue, raising the possibility that both the levels and ratio of intracolonic 25(OH)D and 1,25(OH)₂D concentrations can be modulated resulting in diverse, and stage-specific bioeffects.

In addition to multiplicity of adenomas, other features must be considered. Thus far, results for adenoma size remain equivocal, with one study indicating that the relationship between 25(OH)D and incident adenomas was the same for small vs. larger lesions³, a finding similar to that of Fedirko et al.²⁸. In contrast, Wei et al.¹⁰ conducted a meta-analysis of all studies of 25(OH)D and colorectal adenoma risk, and the results suggested that for advanced adenoma, the inverse association was stronger than that observed for overall adenoma risk; while another report indicated that the magnitude of effect for 25(OH)D was greater for adenomas ≥ 5 mm compared to those < 5 mm³⁰. Peters et al.¹³ investigated the association between 25(OH)D and advanced colorectal adenoma, as defined by the presence of an adenoma > 1 cm and/or with high-grade dysplasia and/or with villous histology. They reported a notable inverse association between 25(OH)D and advanced adenoma among women, but not men¹³. The possibility of differential, sex-specific vitamin D actions in the colorectum must therefore be considered.

For colorectal adenoma specifically, there has been some evidence that the association between circulating 25(OH)D and adenoma is stronger among women than men^{2, 13, 14}, possibly via crosstalk between the vitamin D receptor and estrogen receptor, as reported by Cross et al.⁴⁸. However, more recent studies have indicated no gender differences^{28, 30}. Our current results are in agreement with the latter work, demonstrating no heterogeneity of effect in the association between 25(OH)D and adenoma characteristics and recurrence between men and women. It is possible that the previously-observed variation by gender may relate to colorectal sub-site differences in the formation of adenomas between men and women. For example, it has been proposed that distal and proximal colorectal cancers may arise from different biological pathways⁴⁹, and that these regions have several features that

differentiate them from one another. Proximal and distal regions of the colon arise from different embryonic origins⁴⁹, and have differential patterns of genetic and chromosomal instability^{49, 50}. Chromosomal instability tends to occur more often in the distal region while genetic and microsatellite instability is more often found in proximal neoplasia⁵⁰. However, data for an association between 25(OH)D and adenoma by distal vs. proximal location have been varied. Epidemiological studies in which colonoscopies were conducted and that have included analysis of outcomes by colorectal sub-site have reported stronger associations for proximal adenomas^{29, 30}, distal adenomas^{4, 5}, and no difference between the two^{26, 28}. Investigations employing sigmoidoscopy as the screening method, thus including adenomas of the distal colorectum only, have also reported mixed results, with two studies showing no association with 25(OH)D^{2, 3}, and another reporting a significant relationship with risk of advanced distal adenoma¹³. In the current work, no variation in the association between 25(OH)D and adenoma was observed by colorectal sub-site for baseline adenomas or recurrent adenomas.

Strengths of this work include the large study population, which included over 2000 participants with complete data for adenoma characteristics, dietary intake, and measured 25(OH)D concentrations. The availability of data from two separate study populations allowed for consideration of the consistency of associations. Limitations include the observational nature of the study, which precludes drawing conclusions about causality, as well as the use of a single measure of 25(OH)D for assessment of vitamin D status. Future studies of longitudinal 25(OH)D concentrations will aid in clarifying any role of vitamin D in colorectal neoplasia. An additional limitation is that all participants had had at least one adenoma removed at baseline, and as such conclusions cannot be drawn about any associations between vitamin D metabolite levels and the odds of forming incident adenomas.

In summary, in the present study 25(OH)D was significantly inversely associated with the presence three or more adenomas at baseline, as well as with villous histology, but was also significantly directly related to the development of ≥ 3 recurrent lesions. While we cannot exclude the possibility of these results being due to chance, these unexpected findings may arise from unaddressed variables such as genetic contributors to transport of vitamin D metabolites, or efficiency of conversion of 25(OH)D to the active metabolite, 1,25(OH)₂D at the tissue level. Future work will include a detailed analysis of the contributors to circulating concentrations of 1,25(OH)₂D in this population, as well as the degree to which 1,25(OH)₂D may be associated with the development of colorectal adenoma. Further, genetic variation in the vitamin D pathway will be assessed for any modification of the association between vitamin D metabolites and the development of colorectal adenoma.

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References

1. Haussler MR, Whitfield GK, Kaneko I, Haussler CA, Hsieh D, Hsieh JC, Jurutka PW. Molecular mechanisms of vitamin D action. *Calcif Tissue Int.* 2013; 92:77–98. [PubMed: 22782502]
2. Platz EA, Hankinson SE, Hollis BW, Colditz GA, Hunter DJ, Speizer FE, Giovannucci E. Plasma 1,25-dihydroxy- and 25-hydroxyvitamin D and adenomatous polyps of the distal colorectum. *Cancer Epidemiol Biomarkers Prev.* 2000; 9:1059–65. [PubMed: 11045788]

3. Levine AJ, Harper JM, Ervin CM, Chen YH, Harmon E, Xue S, Lee ER, Frankel HD, Haile RW. Serum 25-hydroxyvitamin D, dietary calcium intake, and distal colorectal adenoma risk. *Nutr Cancer*. 2001; 39:35–41. [PubMed: 11588900]
4. Takahashi R, Mizoue T, Otake T, Fukumoto J, Tajima O, Tabata S, Abe H, Ohnaka K, Kono S. Circulating vitamin D and colorectal adenomas in Japanese men. *Cancer Sci*. 2010; 101:1695–700. [PubMed: 20507319]
5. Adams SV, Newcomb PA, Burnett-Hartman AN, White E, Mandelson MT, Potter JD. Circulating 25-hydroxyvitamin-D and risk of colorectal adenomas and hyperplastic polyps. *Nutr Cancer*. 2011; 63:319–26. [PubMed: 21432725]
6. Zheng XE, Lipka S, Li T, Shahzad G, Levine E, Vlacancich R, Takeshige U, Mustacchia P. The relationship of vitamin D status, smoking, and colorectal adenoma: A retrospective study in an ethnically diverse community. *J Steroid Biochem*. 2012 [Epub ahead of print]. 10.1016/j.jsbmb.2012.09.018
7. LePane CA, Singh G, Spanier-Stiasny JA, Svinarich DM, Rasansky RJ, Hoffman SMJ. Implications of serum 25-Hydroxyvitamin D on the prevalence of neoplastic polyps: A cross-sectional study. *Gastroenterology Res*. 2011; 4:43–50.
8. Yin L, Grandi N, Raum E, Haug U, Arndt V, Brenner H. Meta-analysis: Serum vitamin D and colorectal adenoma risk. *Prev Med*. 2011; 53:10–16. [PubMed: 21672549]
9. Gandini S, Boniol M, Haukka J, Byrnes G, Cox B, Sneyd MJ, Mullie P, Autier P. Meta-analysis of observational studies of serum 25-hydroxyvitamin D levels and colorectal, breast and prostate cancer and colorectal adenoma. *Int J Cancer*. 2011; 128:1414–24. [PubMed: 20473927]
10. Wei MY, Garland CF, Gorham ED, Mohr SB, Giovannucci E. Vitamin D and prevention of colorectal adenoma: a meta-analysis. *Cancer Epidemiol Biomarkers Prev*. 2008; 17:2958–69. [PubMed: 18990737]
11. Gorham ED, Garland CF, Garland FC, Grant WB, Mohr SB, Lipkin M, Newmark HL, Giovannucci E, Wei M, Holick MF. Optimal vitamin D status for colorectal cancer prevention: a quantitative meta analysis. *Am J Prev Med*. 2007; 32:210–6. [PubMed: 17296473]
12. Yin L, Grandi N, Raum E, Haug U, Arndt V, Brenner H. Meta-analysis: longitudinal studies of serum vitamin D and colorectal cancer risk. *Aliment Pharmacol Ther*. 2009; 30:113–25. [PubMed: 19392870]
13. Peters U, Hayes RB, Chatterjee N, Shao W, Schoen RE, Pinsky P, Hollis BW, McGlynn KA. Circulating vitamin D metabolites, polymorphism in vitamin D receptor, and colorectal adenoma risk. *Cancer Epidemiol Biomarkers Prev*. 2004; 13:546–52. [PubMed: 15066918]
14. Jacobs ET, Alberts DS, Benuzillo J, Hollis BW, Thompson PA, Martinez ME. Serum 25(OH)D levels, dietary intake of vitamin D, and colorectal adenoma recurrence. *J Steroid Biochem*. 2007; 103:752–6.
15. Peehl DM, Skowronski RJ, Leung GK, Wong ST, Stamey TA, Feldman D. Antiproliferative effects of 1,25-dihydroxyvitamin D3 on primary cultures of human prostatic cells. *Cancer Res*. 1994; 54:805–10. [PubMed: 7508338]
16. Frampton RJ, Omond SA, Eisman JA. Inhibition of human cancer cell growth by 1,25-dihydroxyvitamin D3 metabolites. *Cancer Res*. 1983; 43:4443–7. [PubMed: 6307514]
17. Palmer HG, Gonzalez-Sancho JM, Espada J, Berciano MT, Puig I, Baulida J, Quintanilla M, Cano A, de Herreros AG, Lafarga M, Munoz A. Vitamin D(3) promotes the differentiation of colon carcinoma cells by the induction of E-cadherin and the inhibition of beta-catenin signaling. *J Cell Biol*. 2001; 154:369–87. [PubMed: 11470825]
18. Wong NA, Pignatelli M. Beta-catenin--a linchpin in colorectal carcinogenesis? *Am J Pathol*. 2002; 160:389–401. [PubMed: 11839557]
19. Alberts DS, Martinez ME, Roe DJ, Guillen-Rodriguez JM, Marshall JR, van Leeuwen JB, Reid ME, Ritenbaugh C, Vargas PA, Bhattacharyya AB, Earnest DL, Sampliner RE. Lack of effect of a high-fiber cereal supplement on the recurrence of colorectal adenomas. Phoenix Colon Cancer Prevention Physicians' Network. *N Engl J Med*. 2000; 342:1156–62. [PubMed: 10770980]
20. Alberts DS, Martínez ME, Hess LH, Einspahr JG, Green SB, Bhattacharyya AK, Guillen J, Krutzsch M, Batta AK, Salen G, Fales L, Koonce K, et al. Phase III Trial of Ursodeoxycholic Acid

- to Prevent Colorectal Adenoma Recurrence. *J Natl Cancer Inst.* 2005; 97:846–53. [PubMed: 15928305]
21. Block G, Hartman AM, Naughton D. A reduced dietary questionnaire: Development and validation. *Epidemiology.* 1990; 1:58–64. [PubMed: 2081241]
 22. Ritenbaugh C, Aickin M, Taren D, Teufel N, Graver E, Woolf K, Alberts DS. Use of a food frequency questionnaire to screen for dietary eligibility in a randomized cancer prevention phase III trial. *Cancer Epidemiol Biomarkers Prev.* 1997; 6:347–54. [PubMed: 9149895]
 23. Hollis BW. Quantitation of 25-hydroxyvitamin D and 1,25-dihydroxyvitamin D by radioimmunoassay using radioiodinated tracers. *Methods Enzymol.* 1997; 282:174–86. [PubMed: 9330287]
 24. Malabanan A, Veronikis IE, Holick MF. Redefining vitamin D insufficiency. *Lancet.* 1998; 351:805–6. [PubMed: 9519960]
 25. Hollis BW. Circulating 25-hydroxyvitamin D levels indicative of vitamin D sufficiency: implications for establishing a new effective dietary intake recommendation for vitamin D. *J Nutr.* 2005; 135:317–22. [PubMed: 15671234]
 26. Peters U, McGlynn KA, Chatterjee N, Gunter E, Garcia-Closas M, Rothman N, Sinha R. Vitamin D, calcium, and vitamin D receptor polymorphism in colorectal adenomas. *Cancer Epidemiol Biomarkers Prev.* 2001; 10:1267–74. [PubMed: 11751444]
 27. Miller EA, Keku TO, Satia JA, Martin CF, Galanko JA, Sandler RS. Calcium, dietary, and lifestyle factors in the prevention of colorectal adenomas. *Cancer.* 2007; 109:510–7. [PubMed: 17200965]
 28. Fedirko V, Bostick RM, Goodman M, Flanders WD, Gross MD. Blood 25-hydroxyvitamin D3 concentrations and incident sporadic colorectal adenoma risk: a pooled case-control study. *Am J Epidemiol.* 2010; 172:489–500. [PubMed: 20650953]
 29. Hong SN, Kim JH, Choe WH, Lee SY, Seol DC, Moon HW, Hur M, Yun YM, Sung IK, Park HS, Shim CS. Circulating vitamin D and colorectal adenoma in asymptomatic average-risk individuals who underwent first screening colonoscopy: a case-control study. *Dig Dis Sci.* 57:753–63. [PubMed: 21984438]
 30. Yamaji T, Iwasaki M, Sasazuki S, Sakamoto H, Yoshida T, Tsugane S. Association between plasma 25-hydroxyvitamin D and colorectal adenoma according to dietary calcium intake and vitamin D receptor polymorphism. *Am J Epidemiol.* 2012; 175:236–44. [PubMed: 22193171]
 31. Garland CF, Comstock GW, Garland FC, Helsing KJ, Shaw EK, Gorham ED. Serum 25-hydroxyvitamin D and colon cancer: eight-year prospective study. *Lancet.* 1989; 2:1176–8. [PubMed: 2572900]
 32. Feskanich D, Ma J, Fuchs CS, Kirkner GJ, Hankinson SE, Hollis BW, Giovannucci EL. Plasma vitamin D metabolites and risk of colorectal cancer in women. *Cancer Epidemiol Biomarkers Prev.* 2004; 13:1502–8. [PubMed: 15342452]
 33. Wactawski-Wende J, Kotchen JM, Anderson GL, Assaf AR, Brunner RL, O'Sullivan MJ, Margolis KL, Ockene JK, Phillips L, Potters L, Prentice RL, Robbins J, et al. Calcium plus vitamin D supplementation and the risk of colorectal cancer. *New Engl J Med.* 2006; 354:684–96. [PubMed: 16481636]
 34. Woolcott CG, Wilkens LR, Nomura AM, Horst RL, Goodman MT, Murphy SP, Henderson BE, Kolonel LN, Le Marchand L. Plasma 25-hydroxyvitamin D levels and the risk of colorectal cancer: the multiethnic cohort study. *Cancer Epidemiol Biomarkers Prev.* 19:130–4. [PubMed: 20056631]
 35. Jenab M, Bueno-de-Mesquita HB, Ferrari P, van Duijnhoven FJ, Norat T, Pischon T, Jansen EH, Slimani N, Byrnes G, Rinaldi S, Tjønneland A, Olsen A, et al. Association between pre-diagnostic circulating vitamin D concentration and risk of colorectal cancer in European populations: a nested case-control study. *BMJ.* 2010; 340:b5500. [PubMed: 20093284]
 36. Tangrea J, Helzlsouer K, Pietinen P, Taylor P, Hollis B, Virtamo J, Albanes D. Serum levels of vitamin D metabolites and the subsequent risk of colon and rectal cancer in Finnish men. *Cancer Causes Control.* 1997; 8:615–25. [PubMed: 9242478]
 37. Wu K, Feskanich D, Fuchs CS, Willett WC, Hollis BW, Giovannucci EL. A nested case control study of plasma 25-hydroxyvitamin D concentrations and risk of colorectal cancer. *J Natl Cancer Inst.* 2007; 99:1120–9. [PubMed: 17623801]

38. Lee JE, Li H, Chan AT, Hollis BW, Lee IM, Stampfer MJ, Wu K, Giovannucci E, Ma J. Circulating levels of vitamin D and colon and rectal cancer: the Physicians' Health Study and a meta-analysis of prospective studies. *Cancer Prev Res.* 2011; 4:735–43.
39. Braun MM, Helzlsouer KJ, Hollis BW, Comstock GW. Colon cancer and serum vitamin D metabolite levels 10–17 years prior to diagnosis. *Am J Epidemiol.* 1995; 142:608–11. [PubMed: 7653469]
40. Otani T, Iwasaki M, Sasazuki S, Inoue M, Tsugane S. Plasma vitamin D and risk of colorectal cancer: the Japan Public Health Center-Based Prospective Study. *Br J Cancer.* 2007; 97:446–51. [PubMed: 17622244]
41. Shabahang M, Buras RR, Davoodi F, Schumaker LM, Nauta RJ, Evans SR. 1,25-Dihydroxyvitamin D3 receptor as a marker of human colon carcinoma cell line differentiation and growth inhibition. *Cancer Res.* 1993; 53:3712–8. [PubMed: 8393379]
42. Wang QM, Jones JB, Studzinski GP. Cyclin-dependent kinase inhibitor p27 as a mediator of the G1-S phase block induced by 1,25-dihydroxyvitamin D3 in HL60 cells. *Cancer Res.* 1996; 56:264–7. [PubMed: 8542578]
43. Holt PR, Arber N, Halmos B, Forde K, Kissileff H, McGlynn KA, Moss SF, Kurihara N, Fan K, Yang K, Lipkin M. Colonic epithelial cell proliferation decreases with increasing levels of serum 25-hydroxy vitamin D. *Cancer Epidemiol Biomarkers Prev.* 2002; 11:113–9. [PubMed: 11815408]
44. Cross HS, Bareis P, Hofer H, Bischof MG, Bajna E, Kriwanek S, Bonner E, Peterlik M. 25-Hydroxyvitamin D(3)-1alpha-hydroxylase and vitamin D receptor gene expression in human colonic mucosa is elevated during early cancerogenesis. *Steroids.* 2001; 66:287–92. [PubMed: 11179736]
45. Matusiak D, Benya RV. CYP27A1 and CYP24 expression as a function of malignant transformation in the colon. *J Histochem Cytochem.* 2007; 55:1257–64. [PubMed: 17875655]
46. Cross HS, Nittke T, Kallay E. Colonic vitamin D metabolism: implications for the pathogenesis of inflammatory bowel disease and colorectal cancer. *Mol Cell Endocrinol.* 2011; 347:70–9. [PubMed: 21801808]
47. Lao VV, Grady WM. Epigenetics and colorectal cancer. *Nature Rev Gastroenterol Hepatol.* 2011; 8:686–700.
48. Cross HS, Kallay E, Lechner D, Gerdenitsch W, Adlercreutz H, Armbrrecht HJ. Phytoestrogens and vitamin D metabolism: a new concept for the prevention and therapy of colorectal, prostate, and mammary carcinomas. *J Nutr.* 2004; 134:1207S–12S. [PubMed: 15113973]
49. Bufill JA. Colorectal cancer: evidence for distinct genetic categories based on proximal or distal tumor location. *Ann Int Med.* 1990; 113:779–88. [PubMed: 2240880]
50. Gervaz P, Bucher P, Morel P. Two colons-two cancers: paradigm shift and clinical implications. *J Surg Oncol.* 2004; 88:261–6. [PubMed: 15565587]

Impact

This work is the largest study conducted to date for vitamin D and colorectal adenoma, and demonstrates an inverse relationship between 25(OH)D concentrations and advanced adenoma features. In addition, the results are consistent with other work showing no association with adenoma recurrence.

Table 1

Baseline characteristics of study participants in the WBF and UDCA Trials

Baseline Characteristics	Pooled	WBF	UDCA	p-value ^{1,2}
	N=2074	N=923	N=1151	
Age at baseline (median, IQR ¹)	67.0 (60.0–72.0)	67.0 (59.0–72.0)	68.0 (61.0–72.0)	0.04
Body mass index (kg/m ² , median, IQR ¹)	27.3 (24.7–30.4)	27.1 (24.5–30.1)	27.5 (24.8–30.8)	0.01
Male, n (%)	1414 (68.2)	640 (69.3)	774 (67.3)	0.31
White, n (%)	1947 (94.9)	877 (95.2)	1070 (94.6)	0.53
Family History CRC, n (%) ³	457 (22.0)	142 (15.4)	315 (27.4)	0.001
Current Smoker (yes), n (%)	263 (12.7)	129 (14.0)	134 (11.6)	0.11
Previous Polyps (yes), n (%) ⁴	826 (39.8)	311 (33.7)	515 (44.7)	0.001
Aspirin Use (yes), n (%) ⁵	580 (28.0)	261 (28.3)	319 (27.7)	0.80

¹Statistical tests for continuous variables comparing participants in WBF to participants in UDCA performed with a Mann-Whitney U-test. Age and BMI were not normally distributed; the medians and interquartile ranges are presented for these variables.

²Statistical tests for categorical variables comparing participants in WBF to participants in UDCA performed with chi-square analyses.

³History of colorectal cancer in one or more first degree relatives.

⁴History of polyps prior to baseline; percentage does not add up to 100 due to missing data (n=167).

⁵Aspirin use in the last month at baseline.

Table 2

Baseline characteristics of participants in the WBF and UDCA trials combined, by 25(OH)D concentration.

Baseline Characteristics	Category of 25(OH)D Concentration (ng/ml)			p-trend ¹
	<20 n=427	≥ 20 and < 30 n=874	≥ 30 n=773	
Age at baseline (median, IQR ²)	66.0 (60.0–72.0)	67.0 (61.0–72.0)	68.0 (61.0–72.0)	0.15
Body mass index (kg/m ² , median, IQR ²)	28.3 (25.0–32.4)	27.6 (24.9–30.7)	26.7 (24.4–29.4)	0.001
Male, n (%)	194 (45.4)	593 (67.9)	627 (81.1)	0.001
White, n (%)	380 (89.6)	817 (94.8)	750 (97.9)	0.001
Family History CRC, n (%) ³	94 (22.0)	211 (24.1)	152 (19.7)	0.15
Current Smoker (yes), n (%)	68 (15.9)	100 (11.4)	95 (12.3)	0.13
Previous Polyps (yes), n (%) ⁴	154 (36.1)	347 (39.7)	325 (42.0)	0.09
Aspirin Use (yes), n (%) ⁵	100 (23.4)	242 (27.7)	238 (30.8)	0.01
Dietary intake				
Energy (kcal/d)	1833.0 ± 801.3	1947.1 ± 755.4	2062.2 ± 742.0	0.001
Fat (g/d)	61.9 ± 33.6	64.8 ± 30.3	69.1 ± 31.0	0.001
Fiber (g/d)	21.1 ± 10.7	22.1 ± 10.5	22.6 ± 9.9	0.05
Vitamin D (IU/d)	107.0 ± 86.5	132.5 ± 97.2	146.1 ± 97.2	0.001
Calcium (mg/d)	849.2 ± 412.7	963.3 ± 464.4	1013.0 ± 441.6	0.001
Folate (mcg/d)	376.8 ± 202.0	406.3 ± 212.0	423.1 ± 212.2	0.001
Magnesium (mg/d)	314.5 ± 128.5	341.5 ± 133.6	355.9 ± 125.8	0.001
Supplement use (yes)	253 (59.3)	577 (66.0)	494 (63.9)	0.22

¹ P-trend calculated with regression modeling using the categorical variable for 25(OH)D status as the independent variable.² Age and BMI were not normally distributed; the medians and interquartile ranges are presented for these variables³ Family history of colorectal cancer in one or more first degree relatives.⁴ History of polyps prior to baseline. Percentages do not add up to 100 due to missing data (n= 167).⁵ Aspirin use in the last month at baseline.

Table 3

Adjusted odds ratios¹ (95% confidence intervals) for the association between 25(OH)D concentration and baseline colorectal adenoma characteristics.

Category of 25(OH)D Concentration	Baseline Adenoma Characteristics (OR, 95% CI)					
	Multiplicity (≥ 3 adenoma)		Large size (≥ 1 cm)		Villous histology ²	
	n (%)	OR (95% CI)	n (%)	OR (95% CI)	n (%)	OR (95% CI)
Pooled population						
<20 ng/ml	63 (14.8)	1.00	180 (42.2)	1.00	101 (23.7)	1.00
≥20 and < 30 ng/ml	156 (17.9)	0.99 (0.70–1.41)	351 (40.2)	0.96 (0.75–1.22)	186 (21.3)	0.88 (0.66–1.18)
≥ 30 ng/ml	122 (15.8)	0.73 (0.50–1.06)	282 (36.5)	0.83 (0.64–1.08)	140 (18.1)	0.72 (0.53–0.99)
p-trend		0.05		0.11		0.04
UDCA trial						
<20 ng/ml	27 (10.7)	1.00	117 (46.3)	1.00	63 (24.9)	1.00
≥20 and < 30 ng/ml	71 (14.5)	1.02 (0.61–1.70)	216 (44.0)	0.94 (0.68–1.30)	102 (20.8)	0.81 (0.56–1.19)
≥ 30 ng/ml	47 (11.6)	0.71 (0.61–1.70)	151 (37.2)	0.70 (0.50–1.00)	71 (17.5)	0.64 (0.42–0.98)
p-trend		0.15		0.03		0.04
WBF trial						
<20 ng/ml	36 (20.7)	1.00	63 (36.2)	1.00	38 (21.8)	1.00
≥20 and < 30 ng/ml	85 (22.3)	1.00 (0.62–1.62)	135 (35.3)	0.99 (0.67–1.45)	84 (22.0)	1.01 (0.65–1.59)
≥ 30 ng/ml	75 (20.4)	0.74 (0.45–1.23)	131 (35.7)	1.01 (0.68–1.45)	69 (18.8)	0.86 (0.53–1.38)
p-trend		0.16		0.93		0.44
p-interaction ³		0.99		0.10		0.33
Men						
<20 ng/ml	39 (20.1)	1.00	75 (38.7)	1.00	43 (22.1)	1.00
≥20 and < 30 ng/ml	118 (19.9)	0.86 (0.55–1.34)	239 (40.3)	1.13 (0.81–1.60)	127 (21.4)	0.94 (0.63–1.40)
≥ 30 ng/ml	113 (18.0)	0.75 (0.48–1.34)	232 (37.0)	1.01 (0.72–1.43)	110 (17.5)	0.72 (0.48–1.09)
p-trend		0.18		0.76		0.06
Women						
<20 ng/ml	24 (10.3)	1.00	105 (45.1)	1.00	58 (24.9)	1.00
≥20 and < 30 ng/ml	38 (13.6)	1.30 (0.74–2.29)	112 (40.0)	0.82 (0.57–1.18)	59 (21.1)	0.81 (0.53–1.25)
≥ 30 ng/ml	9 (6.2)	0.46 (0.20–1.05)	50 (34.3)	0.61 (0.39–0.95)	30 (20.7)	0.78 (0.46–1.31)

Category of 25(OH)D Concentration	Baseline Adenoma Characteristics (OR, 95% CI)					
	Multiplicity (≥ 3 adenoma)		Large size (≥1 cm)		Villous histology ²	
	n (%)	OR (95% CI)	n (%)	OR (95% CI)	n (%)	OR (95% CI)
p-trend		0.13		0.03		0.30
p-interaction ⁴		0.78		0.11		0.84

¹Odds ratios adjusted for body mass index, gender (except for stratified analysis), age, race, and study (for pooled analysis only).

²Villous histology was present if adenoma exhibited tubulovillous or villous histology (at least 25% villous).

³P-interaction for study and vitamin D status calculated using a likelihood ratio test.

⁴P-interaction for gender and vitamin D status calculated using a likelihood ratio test.

Table 4

Adjusted odds ratios¹ (95% confidence intervals) for the association between 25(OH)D concentration and colorectal adenoma recurrence.

Category of 25(OH)D Concentration	Adenoma Recurrence (OR, 95% CI)					
	Any recurrence		Multiplicity (≥3 adenoma)		Large size (≥1 cm)	
	n (%)	OR (95% CI)	n (%)	OR (95% CI)	n (%)	OR (95% CI)
Pooled population						
<20 ng/ml	193 (45.3)	1.00	35 (8.2)	1.00	37 (8.7)	1.00
≥20 and < 30 ng/ml	388 (44.4)	0.91 (0.71–1.17)	110 (12.6)	1.45 (0.96–2.20)	77 (8.8)	0.97 (0.63–1.48)
≥ 30 ng/ml	361 (46.7)	0.95 (0.73–1.24)	118 (15.3)	1.68 (1.09–2.58)	85 (11.0)	1.18 (0.76–1.83)
p-trend		0.85		0.02		0.35
UDCA trial						
<20 ng/ml	105 (51.4)	1.00	14 (5.5)	1.00	22 (8.7)	1.00
≥20 and < 30 ng/ml	207 (42.1)	1.02 (0.73–1.41)	39 (7.9)	1.35 (0.70–2.63)	42 (8.5)	0.95 (0.54–1.69)
≥ 30 ng/ml	173 (42.6)	1.03 (0.72–1.46)	46 (11.3)	1.87 (0.94–3.69)	44 (10.8)	1.18 (0.65–2.14)
p-trend		0.89		0.05		0.48
WBF trial						
<20 ng/ml	88 (50.9)	1.00	21 (12.1)	1.00	15 (8.7)	1.00
≥20 and < 30 ng/ml	181 (47.5)	0.80 (0.55–1.17)	71 (18.6)	1.58 (0.92–2.71)	35 (9.2)	1.02 (0.53–1.96)
≥ 30 ng/ml	188 (51.2)	0.88 (0.60–1.31)	72 (19.2)	1.60 (0.92–2.79)	41 (11.2)	1.21 (0.62–2.34)
p-trend		0.71		0.16		0.50
p-interaction ³		0.91		0.53		0.91
Men						
<20 ng/ml	102 (52.9)	1.00	17 (8.8)	1.00	17 (8.8)	1.00
≥20 and < 30 ng/ml	277 (46.8)	0.77 (0.55–1.08)	83 (14.0)	1.70 (0.97–2.98)	56 (9.5)	1.11 (0.62–2.01)
≥ 30 ng/ml	306 (48.8)	0.86 (0.61–1.20)	106 (16.9)	2.09 (1.19–3.67)	73 (11.6)	1.43 (0.80–2.57)
p-trend		0.72		0.01		0.13
Women						
<20 ng/ml	91 (39.1)	1.00	18 (7.7)	1.00	20 (8.6)	1.00
≥20 and < 30 ng/ml	111 (39.5)	1.11 (0.77–1.61)	27 (9.6)	1.24 (0.65–2.36)	21 (7.5)	0.82 (0.43–1.58)
≥ 30 ng/ml	55 (37.7)	1.02 (0.65–1.60)	12 (8.2)	1.01 (0.46–2.25)	12 (8.2)	0.78 (0.35–1.73)
p-trend		0.87		0.90		0.51
						0.94

Category of 25(OH)D Concentration	Adenoma Recurrence (OR, 95% CI)					
	Any recurrence		Multiplicity (≥3 adenoma)		Large size (≥1 cm)	
	n (%)	OR (95% CI)	n (%)	OR (95% CI)	n (%)	OR (95% CI)
p-interaction ⁴		0.74		0.34		0.23
						1.00

¹ Odds ratios adjusted for body mass index, gender (except for stratified analysis), age, race, and study (for pooled analysis only).

² Villous histology was present if adenoma exhibited tubulovillous or villous histology (at least 25% villous).

³ P-interaction for study and vitamin D status calculated using a likelihood ratio test.

⁴ P-interaction for gender and vitamin D status calculated using a likelihood ratio test.

Table 5

Adjusted odds ratios^I (95% confidence intervals) for the association between 25(OH)D concentration and adenoma, by colorectal sub-site.

Category of 25(OH)D Concentration	Colorectal Sub-site (OR, 95% CI)			
	Distal Adenoma		Proximal Adenoma	
	n (%)	OR (95% CI)	n (%)	OR (95% CI)
Baseline adenomas				
<20 ng/ml	298 (72.2)	1.00	211 (51.1)	1.00
≥20 and < 30 ng/ml	568 (66.4)	0.71 (0.55–0.94)	464 (54.2)	1.11 (0.86–1.42)
≥ 30 ng/ml	520 (68.7)	0.78 (0.59–1.04)	382 (50.5)	0.94 (0.72–1.22)
p-trend		0.21		0.43
Recurrent adenomas				
<20 ng/ml	92 (22.0)	1.00	134 (32.0)	1.00
≥20 and < 30 ng/ml	209 (24.0)	1.03 (0.77–1.38)	289 (33.2)	1.04 (0.80–1.35)
≥ 30 ng/ml	201 (26.1)	1.12 (0.83–1.52)	270 (35.1)	1.10 (0.83–1.45)
p-trend		0.41		0.49

^I Odds ratios adjusted for body mass index, gender, age, race, and study.